INCH-POUND

MIL-F-70546 (AR)
AMENDMENT 2
25 January 2001
SUPERSEDING
AMENDMENT 1
28 September 1994

MILITARY SPECIFICATION

FIN ASSEMBLIES, MORTAR CARTRIDGE, 81MM, M28, M29

MIL-F-70546 was inactivated after 4 January 1997 for new design.

This amendment forms a part of MIL-F-70546 (AR), dated 5 September 1986, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.2, Following E-8 Tension Testing of Metallic Materials:

Add: "E-340 Macroetching Metals and Alloys"
* Add: "ASTM B-117 Standard Test Method of Salt Spray (Fog) Testing"

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2.3, Third line, delete "detain" and substitute "detail"

3.4, Delete "housing" and "fin housing" and substitute "cartridge housing" wherever it appears in the specification.

* Add new paragraph 3.8 as follows:

"3.8 Salt spray test. After anodizing, fins and housings shall be able to withstand salt spray testing in accordance with 4.5.6."

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4.3.1, Add the following to “First article part description”:

“Tensile specimen from cartridge housing stock	See ASTM E-8	2
Tensile specimen from fin stock	See ASTM E-8	2
Microstructure specimen from cartridge housing tensile specimen	---	2
Microstructure specimen from fin stock tensile specimen	---	2
Macroetch specimen from fin stock	---	1”

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Under Fin, after Tension Test, add the following:

	“Macrostructure examination	1 1/	100%	3.3	4.5.6/Visual
	Microstructure examination	2	100%	3.3	4.5.7/Visual
*	Salt spray test	2 2/	( )	3.2	4.5.8”

Under Housing, Cartridge, after Tension Test, add the following:

	“Microstructure examination	2	100%	3.3	4.5.7/Visual
*	Salt spray test	2	( )	3.2	4.5.8”

Under Notes, add the following:

“1/ Examination performed on bar stock.

* 2/ Specimens for this test may be taken from the samples used for the examination for defects, or separate components may be used.”

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4.4.2.1:

Major 101, under sample units, add “1/”

Under AQL, delete “100%”

Add Major 102 as follows:

“102	Microstructure examination	2/	(-)	3.3	4.5.7/Visual”

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Under Notes, add:

"1/ See 4.5.5.1 All specimens must pass.
2/ One per tension test specimen."

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* 4.4.2.2, Major 118, delete examination callout in its entirety and substitute the following:

"118 True position of rear thread PD with ignition cartridge cavity"

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4.4.2.2, Major 130, delete in its entirety.

* 4.4.2.2:

Add Major defects 131, 132 and 133 as follows:

"131	Coating missing or inadequate	( )	0.40%	3.2	Visual
132	Presence of coating on knurl	( )	0.40%	3.2	Visual
133	Salt spray test	1/	( )	3.2	Visual"

Add Minor 205 as follows:

"205	Marking missing, incomplete or illegible	( )	0.65%	3.2	Visual"

Add Note 1 as follows:

"1/ See 4.4.3.1."

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4.4.2.3:

Drawing Number Box, after "9327834" add "9381092"

Major 106, delete "rear chamber" and substitute "start of rear OD"

* Major 107, delete "fo counterbore at rear"

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4.4.2.3:

Major 119, under sample units, add "1/"

Under AQL, delete "100%"

Add Major defect 120 as follows:

"120 Microstructure examination 2/ (-) 3.3 4.5.7/Visual"

* Add Major defects 121, 122, 123 and 124 as follows:

"121	Coating missing or inadequate	( )	0.40%	3.2	Visual
122	Presence of coating in ID further than .094 inch from front face	( )	0.40%	3.2	Visual
123	Radii (2 places) missing or improper on the face of the 35 degree angle of any blade	( )	0.40%	3.2	Gage
124	Salt spray test	3/	( )	3.2	Visual"

Under Notes add the following:

"1/ See 4.5.5.2. All specimens must pass.
2/ One per tension test specimen
* 3/ See 4.4.3.1."

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* 4.4.3, Add "and the following:" at the end of the sentence.

* 4.4.3.1 Add new paragraph 4.4.3.1 as follows:

"4.4.3.1 Salt spray test. See 4.5.8. For this test, a lot of anodized parts shall be the quantity anodized during 8 hours (max) of production. From each lot of anodized fins or housings, three units shall be randomly selected for the test.

If the fins and housings are part of the same anodizing lot, three units of each type shall be tested. See 6.8. In the event that only one type of component fails the test, only the units of the same type in that lit shall be rejected.

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After 10 consecutive production lots pass the test, sampling may be reduced to twice weekly for 2 weeks, and if successful, further reduction to one test every 2 weeks is authorized. If a failure occurs during any phase of reduced sampling, then the sampling shall revert back to the original plan, until 10 consecutive lots pass, etc."

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4.5.5, After 4.5.5 Tension test, add the following:

"The materials shall be certified from the aluminum vendor as meeting the applicable mechanical properties. In addition, the contractor shall have the following samples tested independently from the aluminum vendor. Samples shall be selected by the contractor after receipt of the certified material."

4.5.5.1, Delete "R" and substitute "round" wherever it appears.

Add new paragraphs 4.5.6 and 4.5.7 as follows:

"4.5.6 Macroetch test. Following nominal cropping fill cross section macroetch specimens shall be obtained from the rear of each extruded length, regardless of extrusion method. Additionally, the front of the first acceptable extruded stock from the billet which represents the top of each ingot ("log") shall be macroetched as above. If the front/rear identify of each billet is not maintained, or if the top billet from an ingot is not distinguished from successive billets from the ingot, then macroetch specimens shall be obtained from both ends of each billet. Defective material shall be discarded until the requirements of 3.1 are met. Macroetching shall be in accordance with ASTM E-340, except sodium hydroxide shall be the reagent used.

4.5.7 Microstructure test. From each of the samples submitted for tension testing, one wafer, approximately one-half inch square, shall be prepared for metallographic examination from an undeformed portion of the specimen. The specimen shall be examined in the direction which is perpendicular to the extrusion flow. When suitably prepared, the specimen shall be etched with Keller's reagent and examined by scanning under a metallurgical microscope at a minimum magnification of 500 diameters. Overheating shall be evident by the presence of rosettes in the microstructure or incipient melting at the grain boundaries. Evidence of the above shall cause rejection of all material represented by the applicable sample."

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* Add new paragraph 4.5.8 as follows:

"4.5.8 Salt spray test. See 4.4.3.1. Unless otherwise specified, equipment and procedures for testing shall conform to ASTM B117. Component parts (not test panels) shall be used. Before testing, the parts should be washed in distilled or deionized water and dried with a soft, clean cloth. The test specimens shall be fixtured between 6 and 30 degrees from vertical during testing. The concentration of the salt spray shall be 5 percent. The duration of the test shall be as specified on the applicable drawing.

After testing, the parts shall be rinsed in clean water and dried by air. The significant surface shall be examined to determine that there are no corrosion spots larger than 0.010 inches in diameter, and that there are no patchy dark gray areas of spots, streaks, or marks. The significant surface shall be the entire area within approximately 45 degrees of the portion of the component which faced straight up in the chamber, excluding any area within 0.062 inches of stamp marks, edges and electrode contact marks. Failure of a sample for any of the above reasons shall be cause for rejection of the entire applicable quantity."

6.1, Delete "M853" and substitute "M853A1" wherever it appears.

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* Add new paragraph 6.8 as follows:

"6.8 Anodizing process. The supplier should be thoroughly familiar with all relevant aspects of MIL-A-8625, the impact of the anodic coating on thread dimensions (which must be checked after anodizing).

Ferrous material (steel wire wheels) should not be used for deburring.

Anodizing fins and housings (or other items) simultaneously (mixed loads) is not recommended due to the possible need to compromise process parameters.

Although test panels are not permitted for salt spray testing, they are acceptable for maintaining control and verifying compliance with other requirements of the process, such as unit coating weight, coating thickness, and so on.

During the anodizing process, it would serve the customer's long term need best if components were fixtured in a way that would minimize or completely prevent the outside surfaces from making any contact with the electrodes/racks, while at the same time permitting easy fluid flow and drainage in the internal area. It is the external surfaces which the most continuous coating possible."

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AMENDMENT 2

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